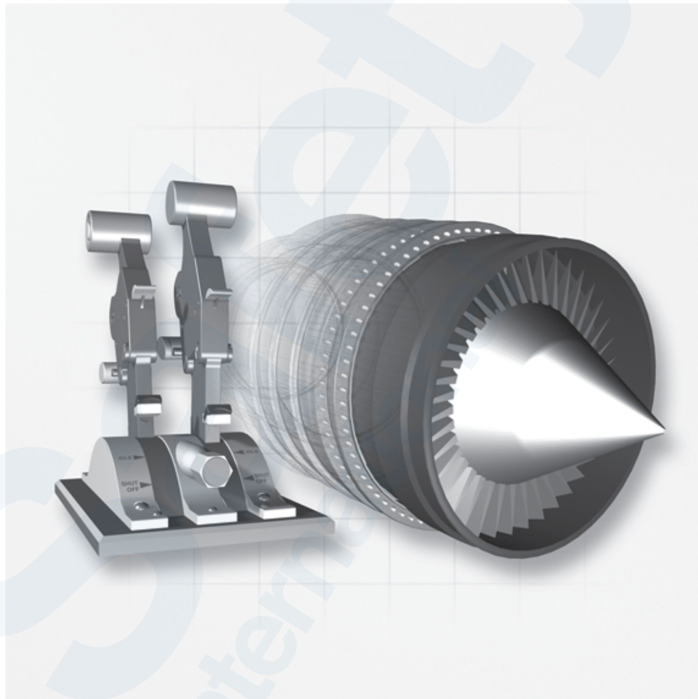
CHAPTER 78
ENGINE EXHAUST



78-00-00 INTRODUCTION

The exhaust system contains the hot engine exhaust gas as it moves rearward from the engine through the nacelle.

GENERAL

The exhaust system is made from the following two assemblies:

- Jet pipe assembly and
- Shroud assembly.

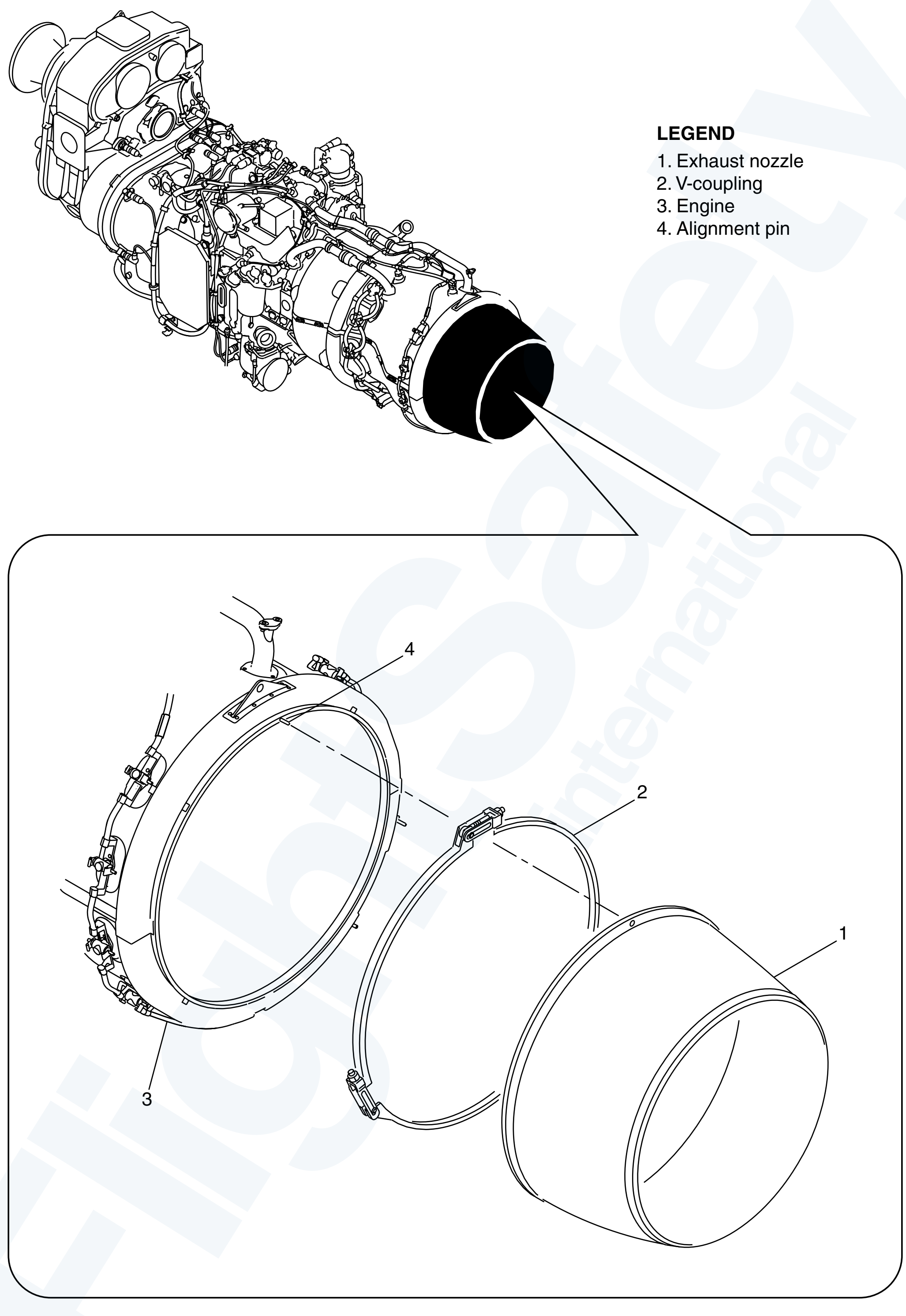


Figure 78-1. Exhaust Nozzle

SYSTEM OVERVIEW

The Jet pipe is routed under the wing box and over the main landing gear bay. The exhaust gas goes through the jet pipe and into the atmosphere at the top rear surface of the nacelle.

The exhaust system has the components that follow:

- Exhaust nozzle
- Exhaust nozzle shroud
- Forward jet pipe
- Aft jet pipe
- Forward shroud
- Mid shroud
- Aft shroud
- Insulation blankets
- Trunnion bearings
- Aft exhaust outlet.

PRINCIPAL COMPONENTS

EXHAUST NOZZLE

Refer to Figure 78-1. Exhaust Nozzle.

The exhaust nozzle is attached to the rear of the engine by a V-coupling.

The exhaust nozzle is a tapered cylindrical design open at the aft end.

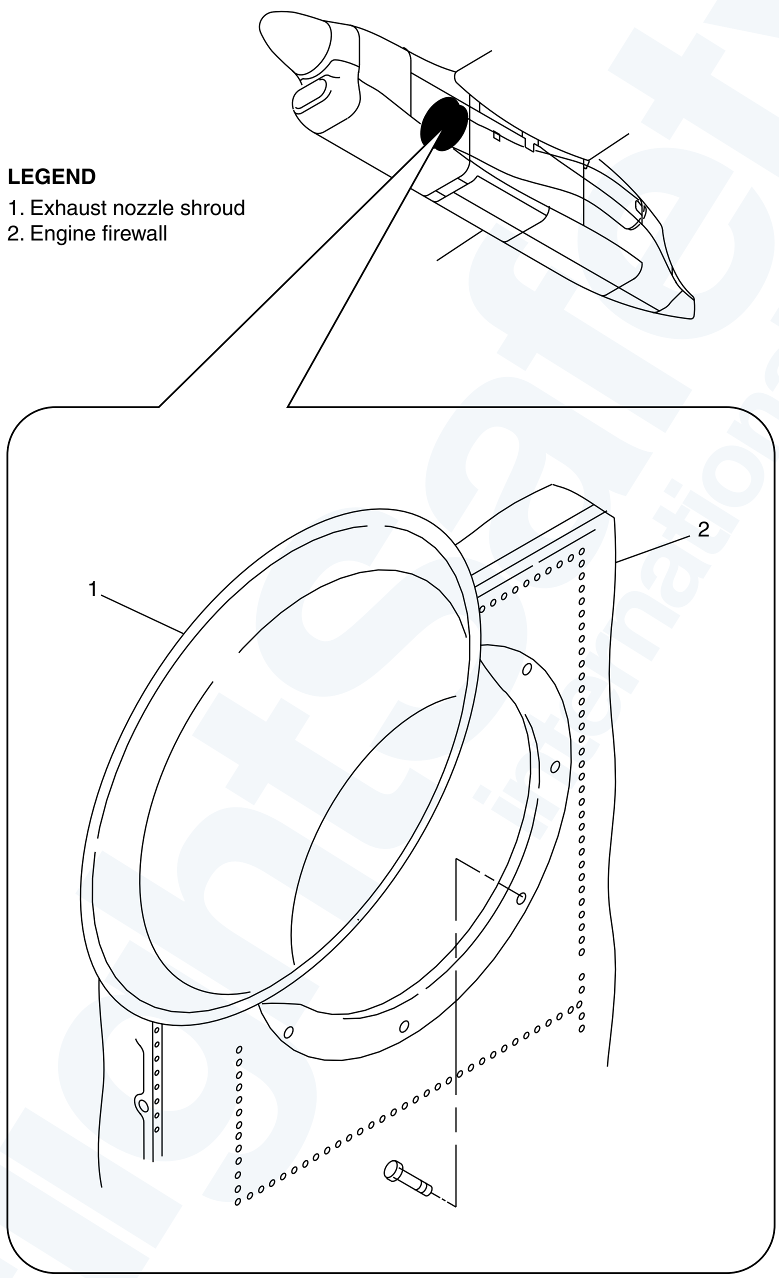


Figure 78-2. Exhaust Nozzle Shroud

EXHAUST NOZZLE SHROUD

NOTES

Refer to:

- Figure 78-2. Exhaust Nozzle Shroud.
- Figure 78-3. Exhaust Bell.

The exhaust nozzle shroud is installed around the exhaust nozzle and operates as a firewall between it and the adjacent area.

A flange, welded to the rear of the exhaust nozzle shroud attaches to the engine firewall.

Figure 78-3. Exhaust Bell

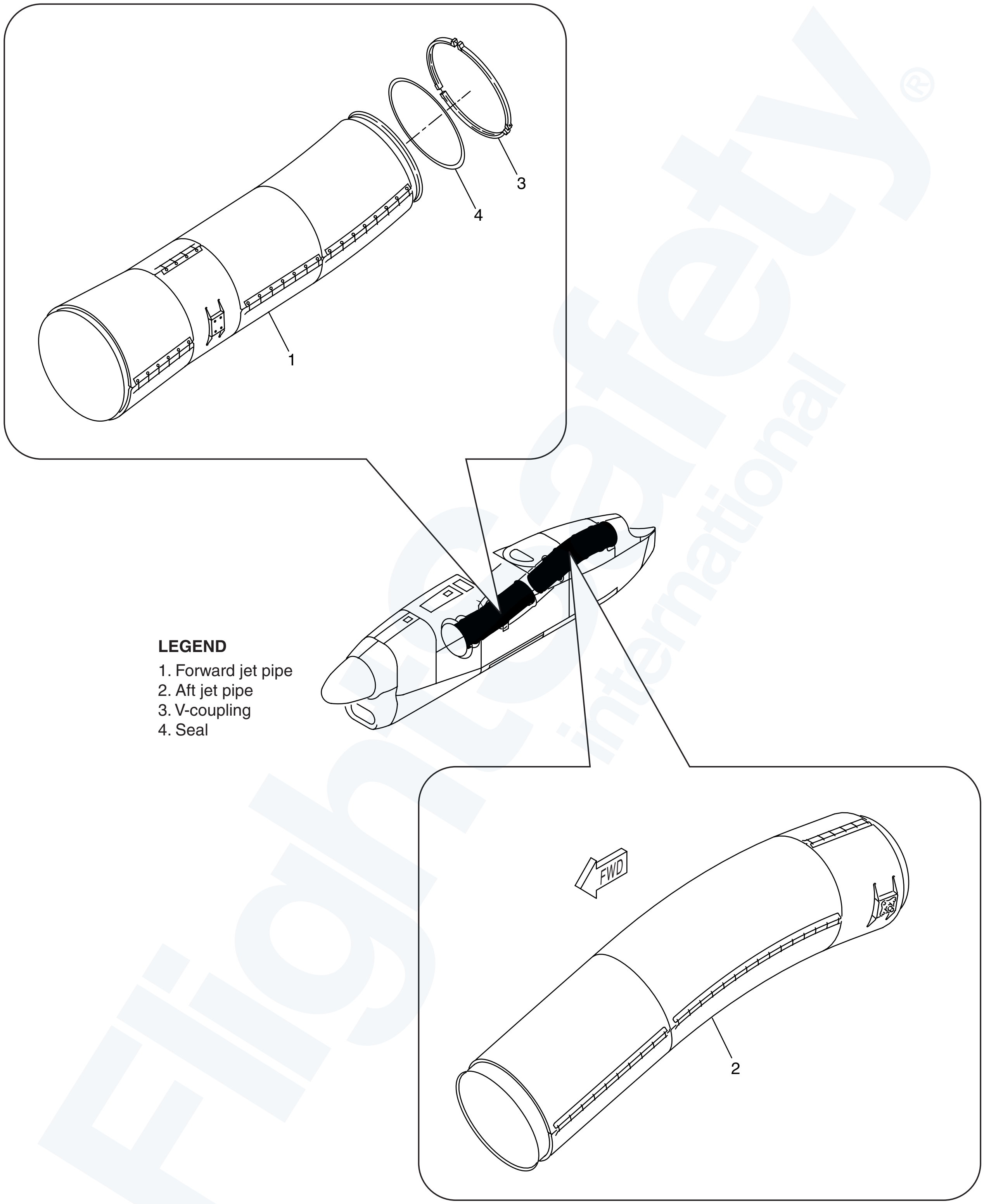


Figure 78-4. Forward and Aft Jet Pipes

FORWARD AND AFT JET PIPES

Refer to:

- Figure 78-4. Forward and Aft Jet Pipes.
- Figure 78-5. Fwd Attachment.

The forward jet pipe is attached to the zone 1 Titanium firewall, in the center of the nacelle, under the wing box. A P-seal and flange seal configuration secures the front of the forward jet pipe in position.

The rear of the forward jet pipe connects with the aft jet pipe in the aft of the nacelle, above the MLG wheel bay. It is connected by an E-seal and V-coupling configuration.

Two forward mounting pin assemblies on each side of the forward jet pipe attach it to the nacelle structure. Two aft mounting pin assemblies have a trunnion bearing, which moves longitudinally in the slot of a bracket which attach it to the nacelle structure.

The jet pipe directs the hot exhaust gases to atmosphere beyond the aircraft structure.

Figure 78-5. Fwd Attachment

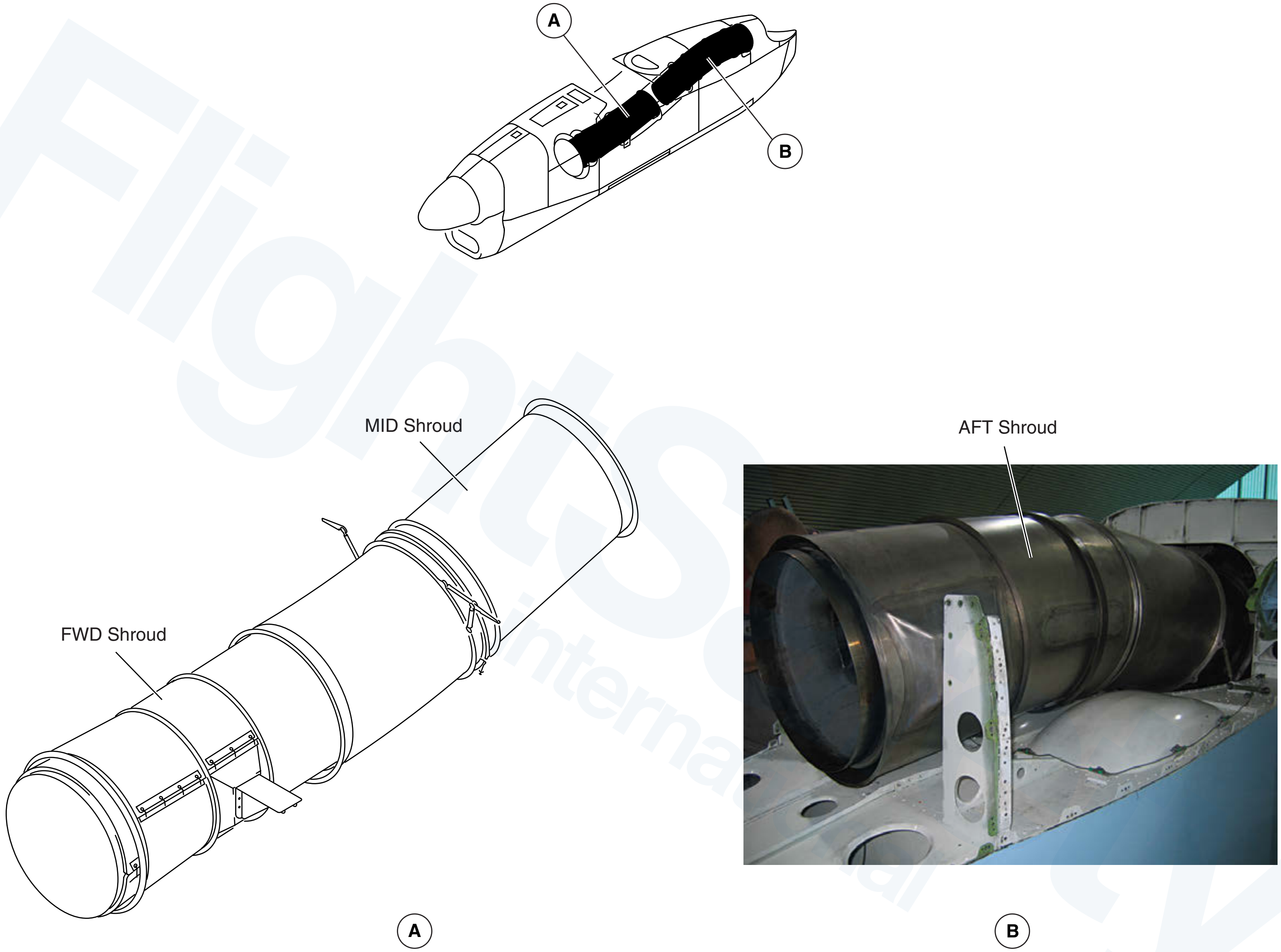


Figure 78-6. Forward Mid and Aft Shrouds

FORWARD, MID AND AFT SHROUDS

Refer to Figure 78-6. Forward Mid and Aft Shrouds.

The forward, mid and aft shrouds are installed around the two jet pipes.

The forward shroud is attached by two mounting pin assemblies on each side of the forward jet pipe. This holds the front of the forward shroud in position against the Zone 4 firewall by a P-seal and flange seal arrangement. Heat insulation blankets are installed on the top outer surface of the forward shroud assembly.

The mid shroud is installed between the forward and aft jet pipe. A tie rod assembly on each side of the nacelle structure attaches to the mid shroud. A seal on the rear of the forward shroud engages with the mid shroud. The top rear structure of the nacelle makes part of the shroud assembly.

The aft shroud is attached to the mounting pin assembly on each side of the aft jet pipe. A seal on the front of the aft shroud engages with the mid shroud. A V-coupling attaches the rear of the aft shroud to the aft exhaust outlet. A tie rod assembly on each side of the nacelle structure attaches with the aft shroud.

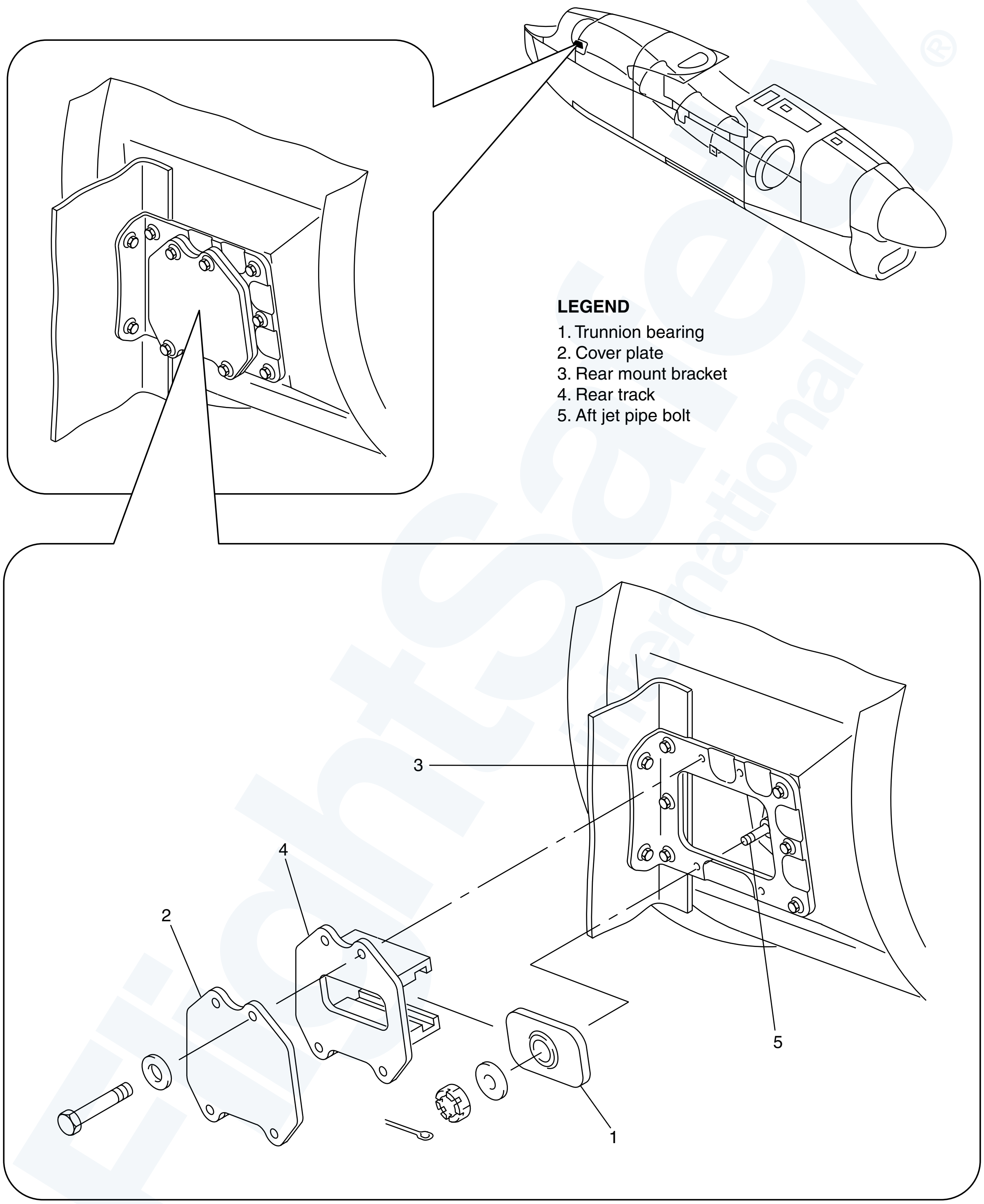
The forward, mid and aft shrouds are each made of titanium. The forward is made from two pieces welded together.

Lockwire attaches the insulation blankets to lugs on the forward shroud. A V-coupling connects the following components:

- Forward shroud to the mid shroud together
- Mid shroud to the aft shroud
- Aft shroud to the aft exhaust outlet.

The shroud assembly is a housing for the jet pipe assembly along its full length.

The shroud is also a firewall for the adjacent area.



LEGEND
1. Trunnion bearing
2. Cover plate
3. Rear mount bracket
4. Rear track
5. Aft jet pipe bolt

Figure 78-7. Trunnion Bearings

TRUNNION BEARINGS

Refer to:

- Figure 78-7. Trunnion Bearings.
- Figure 78-8. Exhaust Pipe.

Trunnion bearings are attached to the two mounting pin assemblies on the Aft jet pipe.

The trunnion bearing moves longitudinally in the slot of a bracket which is installed on the nacelle structure.

The trunnion bearing is a rectangular metal block with a bearing installed in the center.

The trunnion bearings allow the jet pipe to expand and contract with temperature.

NOTES

Figure 78-8. Exhaust Pipe

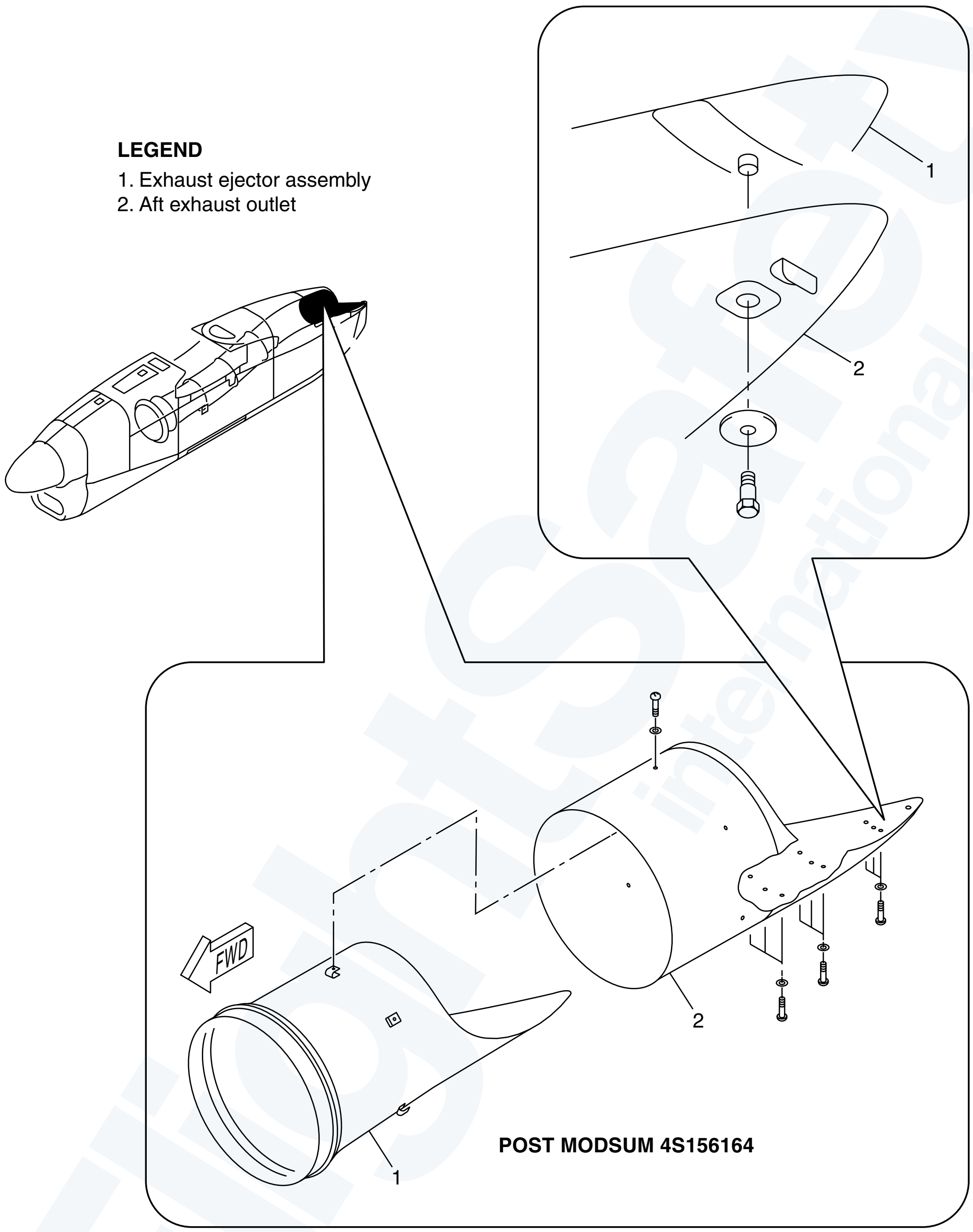


Figure 78-9. Aft Exhaust Outlet

AFT EXHAUST OUTLET

Refer to:

- Figure 78-9. Aft Exhaust Outlet.
- Figure 78-10. Aft Exhaust Outlet Photo (1 of 2).
- Figure 78-10. Aft Exhaust Outlet Photo (2 of 2).

The aft exhaust outlet ejector is attached to the aft jet pipe by a V-coupling.

The aft exhaust extension is around the ejector and is attached to the aft shroud and to the nacelle structure by a V-coupling.

The aft exhaust outlet is a titanium ejector with a titanium extension shroud around it. Both have an extended cutout in the shape of a fingernail at the top rear position to let the engine exhaust gas exit in the air.

The aft exhaust outlet allows for a smooth flow of exhaust gases into the air stream.

INSULATION BLANKETS

The whole jet pipe assembly is covered with insulating blankets. There are eight around the forward jet pipe and six around the aft jet pipe. There are two additional blankets installed on the top of the forward shroud.

The blankets are made from kaowool, sandwiched between two thin sheets of stainless steel. When the insulation blankets are assembled, the kaowool is compressed to an approximate thickness of 3/8 in. (9.53 mm). Each insulation blanket is lockwired in position.

The function of the insulation blankets is to lower the rate of heat release from the jet pipe.

Figure 78-10. Aft Exhaust Outlet Photo (1 of 2)

Figure 78-10. Aft Exhaust Outlet Photo (2 of 2)

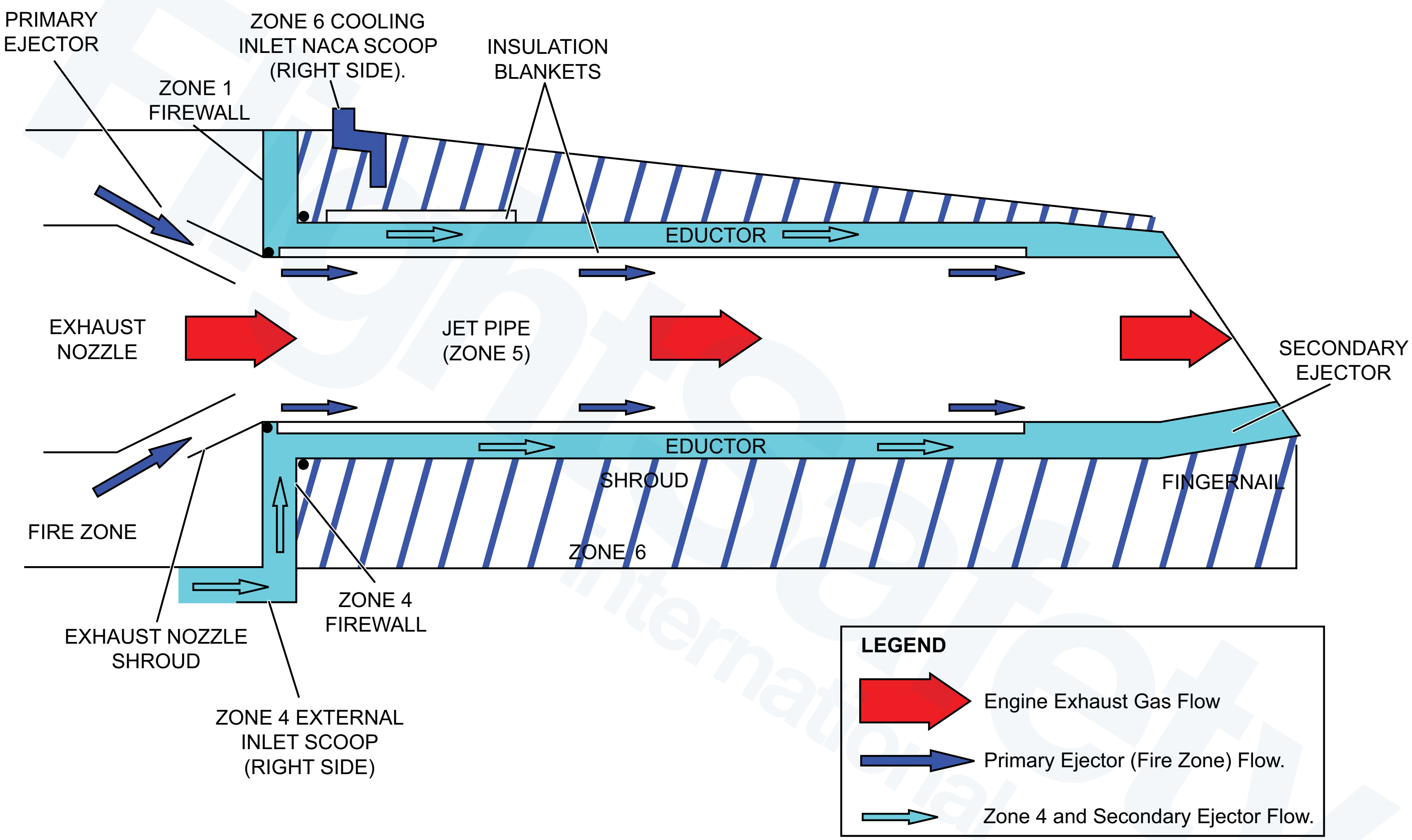


Figure 78-11. Operations (Exhaust Assembly)

OPERATION

NORMAL SYSTEM OPERATION

Refer to:

- Figure 78-11. Operations (Exhaust Assembly).
- Figure 78-12. Exhaust Assembly.

During the engine operation, exhaust flows out the exhaust nozzle into the exhaust nozzle shroud. This will draw air in the engine compartment into the exhaust stream, cooling the exhaust stream. The air in the engine compartment is thereby replaced by cooler air entering the exhaust nozzle. This is referred to as the primary ejector.

The exhaust system is also cooled by an eductor system which allows ambient air to flow between the jet pipe and the shroud.

The air inlet for the eductor system comes from an air inlet scoop on the nacelle RH center side panel. The exhaust gas exiting between the aft exhaust outlet and the aft exhaust outlet shroud creates low pressure which assists in drawing air (between the jet pipe and the shroud) through the eductor, cooling the exhaust system.

This secondary airflow, referred to as the secondary ejector, helps to keep the accessory temperatures in the approved limits.

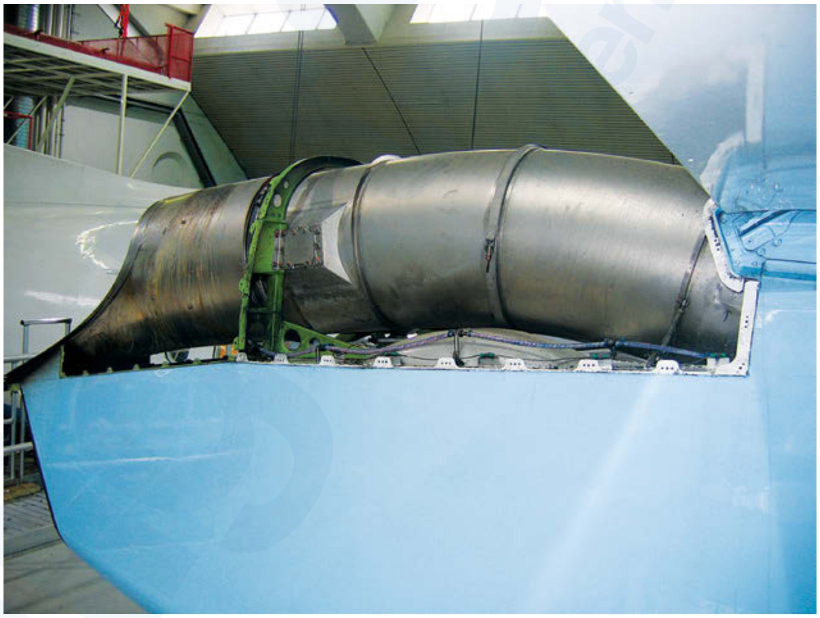


Figure 78-12. Exhaust Assembly

78-00-00 APPENDIX

MAINTENANCE CONSIDERATION

Safety Precautions

WARNING

LET THE EXHAUST SURFACE BECOME COOL BEFORE YOU DO MAINTENANCE. IF YOU DO NOT DO THIS, THE HOT EXHAUST SURFACE WILL CAUSE INJURIES.

YOU MUST INSTALL THE LOCKPINS ON THE MAIN LANDING GEAR. MAKE SURE YOU ENGAGE THE NOSE GEAR LOCK. IF YOU DO NOT DO THIS, THE LANDING GEAR CAN ACCIDENTALLY RETRACT. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

YOU MUST INSTALL THE LOCKPINS IN THE DOOR MECHANISMS OF THE MLG AND NLG. THE DOOR MECHANISMS CAN ACCIDENTALLY CLOSE THE LANDING GEAR DOORS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

CAUTION

INSTALL PACKING BETWEEN THE JET PIPE AND THE SHROUD, AND INSTALL SUPPORTS BELOW THE SHROUD BEFORE THE TIE RODS ARE DISCONNECTED.

IF YOU DO NOT DO THIS, THE EXHAUST ASSEMBLY CAN BE DAMAGED AND CAN CAUSE DAMAGE TO THE NACELLE COMPONENTS.

BE CAREFUL WHEN YOU MOVE THE AFT EXHAUST ASSEMBLY. THE AFT EXHAUST ASSEMBLY COULD HIT THE NACELLE.

78-00-00 SPECIAL TOOLS & TEST EQUIPMENT

- GSB2000001 Borescope - 110 Volt, 60 Hz or equivalent
- GSB2000015 Borescope - 220 Volt, 50 Hz or equivalent

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